Agenda

1. Exam 1 is on February 28, 7:30pm - 9:00pm
    a. Exam Information is on Canvas
    b. Old exams (and solutions) are on canvas
    c. An (optional) WebAssign review assignment is on Canvas
    d. Lots of resources for getting help.
2. Section 7.8 (Improper Integrals)

p-Integrals

Theorem: Let p be a constant.

1.) If $p>1$, then the improper integral $\int_1^\infty \frac{dx}{x^p}$ Converges.

2.) If $p\le 1$, then the improper integral $\int_1^\infty \frac{dx}{x^p}$ diverges.

Proof: we did $p=1$ in L8.

Assume $p \neq 1$,

$$\int_1^\infty \frac{1}{x^p}dx = \lim_{t\to\infty} \int_1^t x^{-p}\,dx = \lim_{t\to\infty}\left[x^{-p+1}\cdot\frac{1}{-p+1}\right]_1^t$$

$$= \lim_{t\to\infty}\left[\frac{t^{-p+1}}{p+1} - \frac{1^{-p+1}}{-p+1}\right]$$

$$= \lim_{t\to\infty}\left[\frac{t^{1-p}}{1-p} - \frac{1}{1-p}\right]$$

If $1-p<0$, $1<p$:

$$= 0 - \frac{1}{1-p} = \frac{-1}{1-p} = \frac{1}{p-1}$$

Some # so it converges

If $1-p>0$, $1>p$:

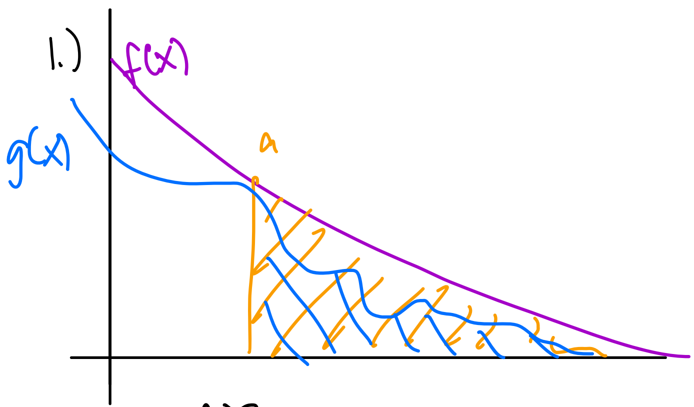
$$= \infty \cdots = \infty$$

Examples: $\int_1^\infty \frac{dx}{x^5}$, $\int_1^\infty \frac{dx}{\sqrt{x}}$, $\int_1^\infty \frac{dx}{x^{1.0001}}$, $\int_1^\infty x^3\,dx$ $\quad \frac{1}{x^{-3}}$

converges $p>1$; diverges $p<1$; converges $p>1$; diverges $p<1$

Comparison Test

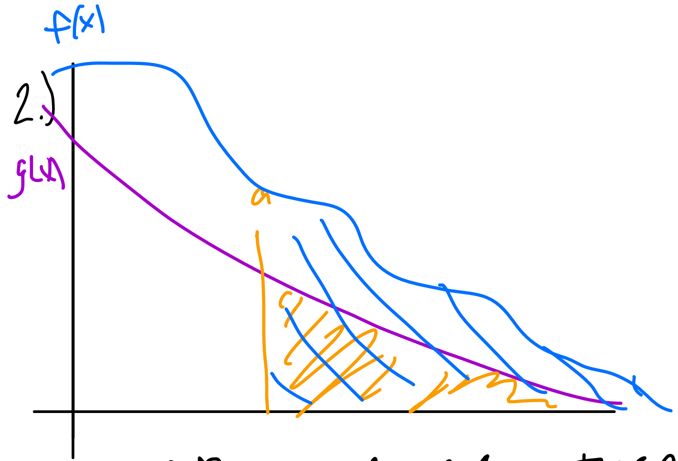
1.)



If $\int_a^\infty f(x)\,dx$ converges,

$\int_a^\infty g(x)\,dx$ converges

2.)



If $\int_a^\infty g(x)\,dx$ diverges, then...

$\int_a^\infty f(x)\,dx$ diverges.

Comparison Test Suppose $f(x)$ and $g(x)$ are continuous on $[a, \infty)$ and $f(x) \geq g(x)$ for all $x \geq a$.

If $\int_a^\infty f(x)\,dx$ converges, then $\int_a^\infty g(x)\,dx$ converges

If $\int_a^\infty g(x)\,dx$ diverges, then $\int_a^\infty f(x)\,dx$ diverges

$1/x^{3/2}$

Example: Determine whether $\int_1^\infty \frac{\cos^2 x}{\sqrt{x^3+1}} \cancel{dx}$ converges or diverges.

1) Pick a $\frac{1}{x^p}$ to compare with:

$1/x^{3/2}$, $p = 3/2 > 1$

note: $\int_1^\infty 1/x^{3/2}\,dx$ converges, then $\int_1^\infty \frac{\cos 2x}{\sqrt{x^3+1}}\,dx$ converges

(big) (small)

2.) Show $1/x^{3/2} \geq \frac{\cos 2x}{(x+1)^{1/2}} \geq 0$

$x \geq 1 \quad x^3 \leq x^3+1$

$\sqrt{x^3} \leq \sqrt{x^3+1}$

$1/x^{3/2} \leq \frac{1}{\sqrt{x^3}} \geq \frac{1}{\sqrt{x^3+1}} \geq \frac{\cos^2 x}{\sqrt{x^2+1}} \geq 0$

Therefore $\int_1^\infty \frac{\cos^2 x}{\sqrt{x^3+1}}$ converges by comparison with $\int_1^\infty \frac{1}{x^{3/2}}\,dx$

Activity: Determine whether $\int_1^\infty \frac{dx}{\sqrt{x}(\sqrt{x}-1)}$ Converges or ~~diverges~~

Take $p=1$:

$\int_2^\infty 1/x\,dx$ diverges

$x - \sqrt{x} \leq x$

$\sqrt{x} \leq 0$

By comparison test, $\int_2^\infty \frac{1}{\sqrt{x}(\sqrt{x}-1)}\,dx$ diverges

b/c $\int_2^\infty 1/x\,dx$ diverges.

Improper Integrals on Discontinuous Integrals

Definition: Let $f(x)$ be a function.

1.) If $f(x)$ is continuous on $[a, b)$ and discontinuous at $x=b$, then

$$\int_a^b f(x)\,dx = \lim_{t\to b^-} \int_a^t f(x)\,dx$$

2.) If $f(x)$ is continuous on $(a, b]$ and discontinuous at $x=a$, then

$$\int_a^b f(x)\,dx = \lim_{t\to a^+} \int_t^b f(x)\,dx$$

Example: $\int_3^7 \frac{dx}{\sqrt{x-3}}$

Activity: $\int_1^2 \frac{4x}{\sqrt[3]{x^2-4}}\,dx$

Example: $\int_{-1}^1 \frac{1}{x}\,dx$